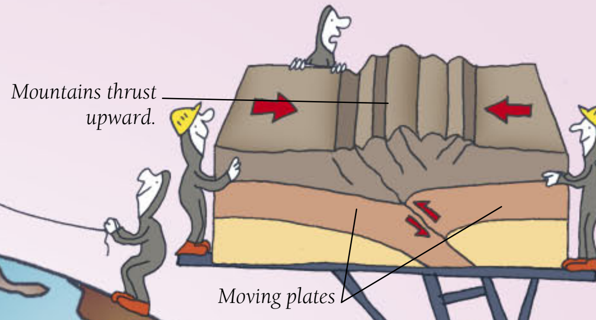
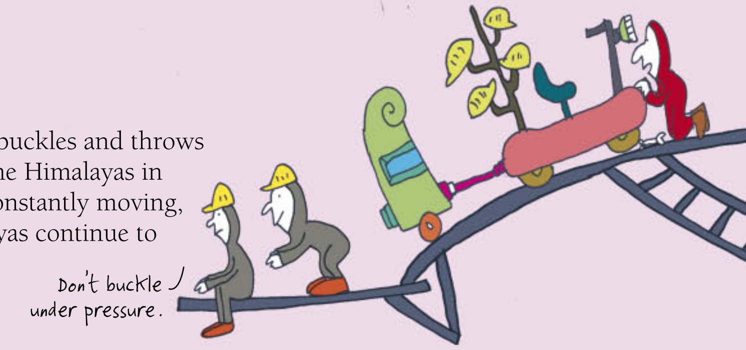
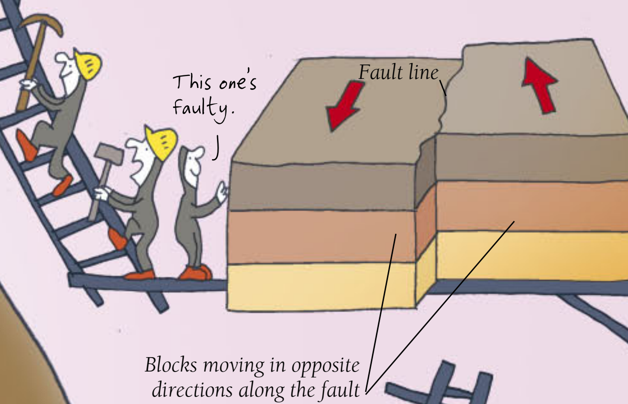


Buckling plates

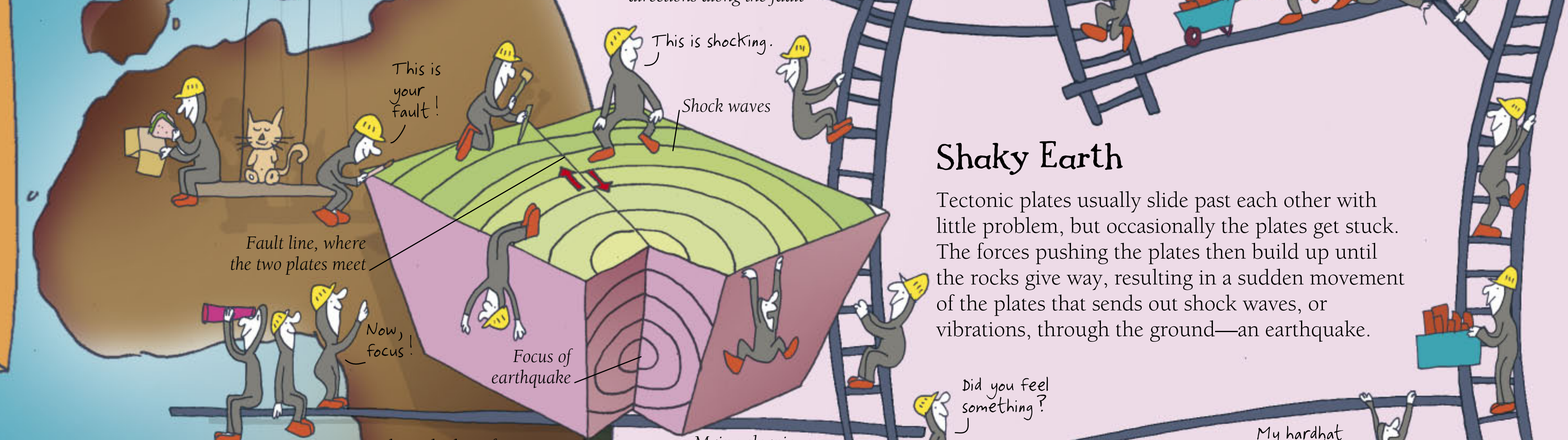
As one tectonic plate hits another, it buckles and throws up a huge mountain chain, such as the Himalayas in central Asia. Because the plates are constantly moving, this process never ends—the Himalayas continue to rise by about 0.2in (5mm) each year.





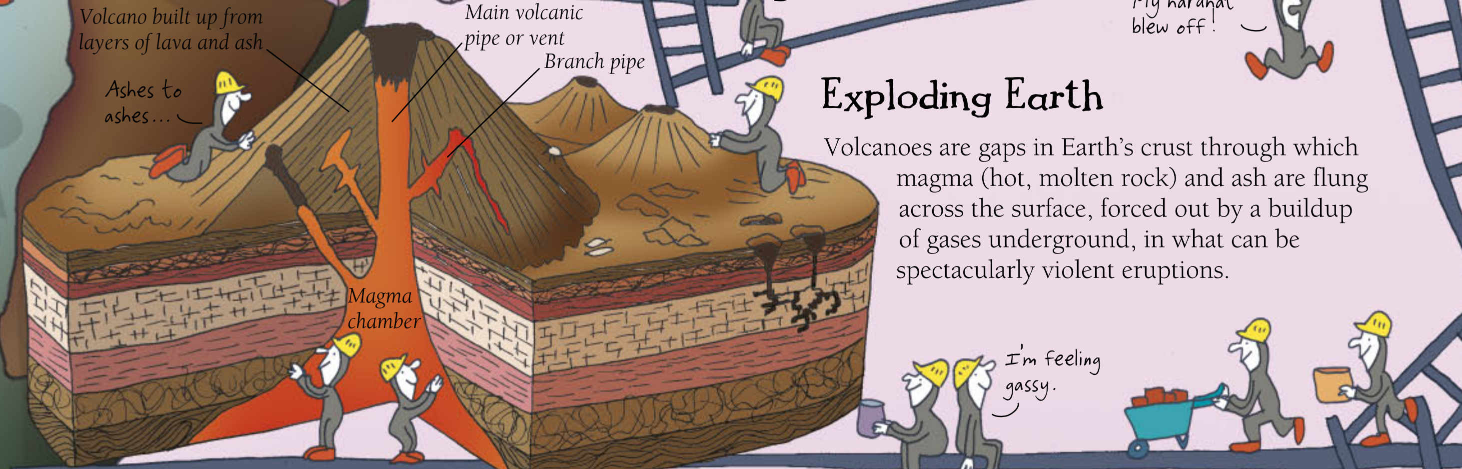
Fractured Earth

As Earth's tectonic plates knock into or slide past each other, they often put the rocks under such strain that they crack or fracture into faults. This allows the blocks of rock to move up, down, or sideways against each other along these faults.



Shaky Earth

Tectonic plates usually slide past each other with little problem, but occasionally the plates get stuck. The forces pushing the plates then build up until the rocks give way, resulting in a sudden movement of the plates that sends out shock waves, or vibrations, through the ground—an earthquake.



Exploding Earth

Volcanoes are gaps in Earth's crust through which magma (hot, molten rock) and ash are flung across the surface, forced out by a buildup of gases underground, in what can be spectacularly violent eruptions.

Rock cycle

As Earth has evolved over millions of years, three main types of rock have formed in its crust: igneous rocks formed as molten magma solidified, metamorphic rocks were transformed by heat or pressure, while sedimentary rocks are compacted debris that settled on the ocean floor millions of years ago.